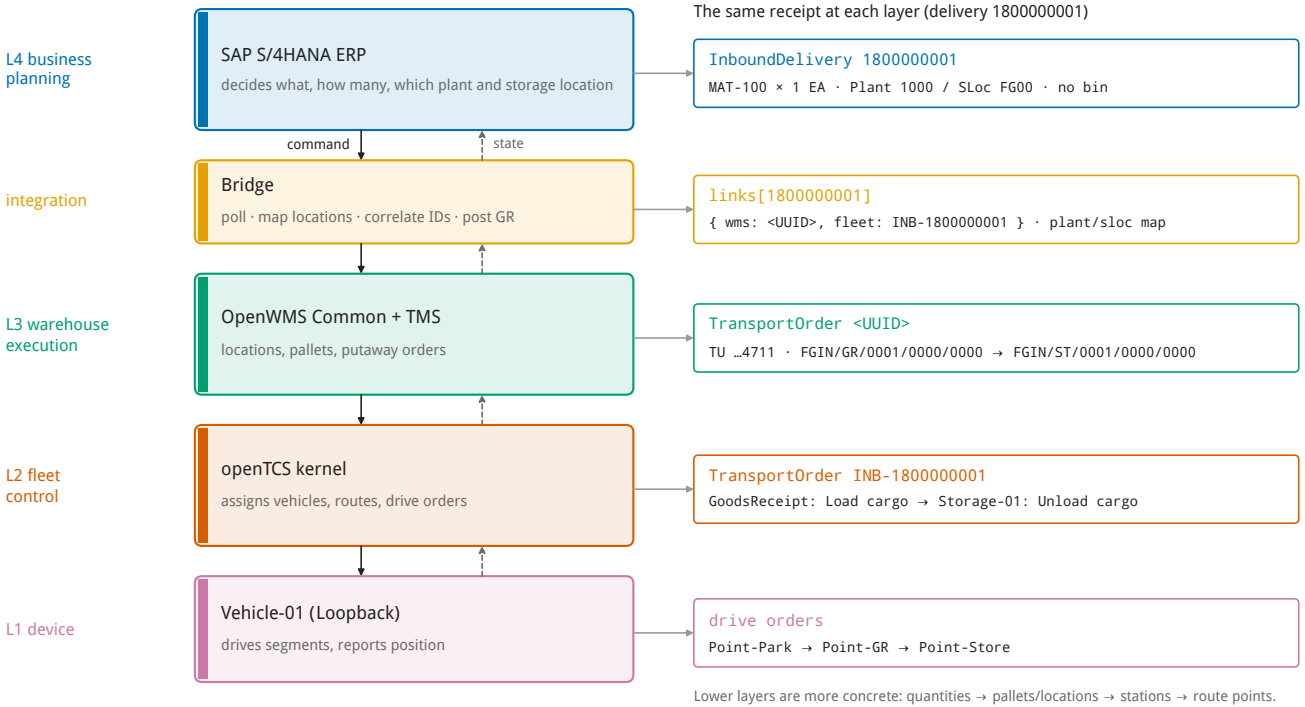
a Layers: commands are refined downwards, states reported upwards



b Two mappings kept by the bridge

Layer	Identifier	Source	Target
SAP	1800000001	1000 / FG00	GR 101
OpenWMS	<UUID, per run>	FGIN/GR/0001/…	FGIN/ST/0001/…
openTCS	INB-1800000001	GoodsReceipt	Storage-01
Vehicle	Vehicle-01	Point-GR	Point-Store

Stage	SAP	OpenWMS	openTCS	Vehicle
waiting	'A'	–	–	IDLE
putaway	'A'	STARTED	BEING_PROCESSED	EXEC.
posted	'C'	FINISHED	FINISHED	IDLE
parked	'A'	ONFAILURE	FAILED	–

The inbound delivery stays 'A' while the vehicle moves, and SAP has no bin. Progress is visible only in WMS and the fleet. Levels follow ISA-95.